Logic, Language, and Philosophy

Predicate Logic: Validity

Meeting 19, November 24, 2025

Martin Stokhof

Department of Philosophy, Tsinghua University, Beijing
ILLC/Department of Philosophy, University of Amsterdam

Overview

Recap: the truth definition

Some facts

Validity

- Formulas
- Argument schemes
- Equivalences

Proving invalidity

Homework

Goals for today

Main goals:

- Illustrate the truth definition by means of some general facts, and illustrate how these can be proven
- Illustrate the concept of validity by showing how to prove validity and how to construct counterexamples

Note:
In what follows we will only use interpretation with assignments

Remark on translation

Note:

- Conditional statements and arguments are related *but not the same thing*

Example:

- If John gets a passing grade, someone is happy
  - $Pj \rightarrow \exists x Hx$
  - $Pj \ / \ \exists x Hx$
- If someone gets a passing grade, they are happy
  - ?? $\exists x Px \ / \ Hx$??
  - $\forall x(Px \rightarrow Hx)$

Exercise revisited

Consider the following model for a predicate logical language with a 1-place predicate letter A, and 2-place predicate letter R:

$$D = \{1, 2, 3, 4\}$$
$$I(A) = \{1, 4\}$$
$$I(R) = \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 1, 3\rangle, \langle 1, 4\rangle, \langle 3, 3\rangle, \langle 4, 1\rangle, \langle 4, 2\rangle, \langle 4, 3\rangle, \langle 4, 4\rangle\}$$

a. Draw a graph of the model
b. Which of the following formulas are true in the model? Motivate your answer.
    i. $\forall x(Ax \leftrightarrow \forall y Rxy)$
    ii. $\forall x \forall y(Rxy \rightarrow Ryx)$
    iii. $\exists x \exists y(Rxy \wedge Ryx)$

Substitutional interpretation revisited

Observation:

- For the sentential part of predicate logic the substitutional interpretation and the assignment method are provably equivalent
- Advantages of the assignment method:
  - also covers (argument schemes with) formulas that are not sentences
  - obeys the compositionality principles in a stricter way
  - does not need to make the language ‘fit’ a model

Question:

- So why consider substitutional interpretation at all?

Substitutional interpretation revisited (2)

- Quine's criterion for ontological commitment: "To be is to be the value of a variable"
- $\rightsquigarrow$ the assignment method ('objectual interpretation')

Claim:

- Substitution interpretation avoids ontological commitments:
- 'There exists a flying horse' is true iff we can find a name *N* s.th. '*N* is a flying horse' is true

But note:

- This shifts the question, does not avoid it:
- Ontological commitments are also a matter of what names *refer to*

Recap: the truth definition

Terms

Let $M = \langle D_M, I_M \rangle$ be a model for L.

Definition (Interpretation of terms)

$$[\![t]\!]_{M,g} = \begin{cases} I_M(t) & \text{if } t \text{ is a constant} \\ g(t) & \text{if } t \text{ is a variable} \end{cases}$$

Truth

Definition (Truth definition)

1. $V_{M,g}(At_1 \dots t_n) = 1$ iff $\langle [\![t_1]\!]_{M,g}, \dots, [\![t_n]\!]_{M,g} \rangle \in I_M(A)$
2. $V_{M,g}(\neg\phi) = 1$ iff $V_{M,g}(\phi) = 0$
3. $V_{M,g}(\phi \wedge \psi) = 1$ iff $V_{M,g}(\phi) = 1$ and $V_{M,g}(\psi) = 1$
4. $V_{M,g}(\phi \vee \psi) = 1$ iff $V_{M,g}(\phi) = 1$ or $V_{M,g}(\psi) = 1$
5. $V_{M,g}(\phi \to \psi) = 1$ iff $V_{M,g}(\phi) = 0$ or $V_{M,g}(\psi) = 1$
6. $V_{M,g}(\forall x\phi) = 1$ iff for all $d \in D_M$: $V_{M,g[x/d]}(\phi) = 1$
7. $V_{M,g}(\exists x\phi) = 1$ iff there is a $d \in D_M$: $V_{M,g[x/d)]}(\phi) = 1$

Some facts

Fact 1

Fact (1)

If $g(v) = g'(v)$ for every variable v that occurs free in ϕ, then:

$$V_{M,g}(\phi) = 1 \textit{ iff } V_{M,g'}(\phi) = 1$$

Proof: by induction on the construction of ϕ.

Proof

Here are three of the cases:

1. $\phi = At_1 \dots t_n$:
   $V_{M,g}(At_1 \dots t_n) = 1$ iff $\langle [\![t_1]\!]_{M,g} \dots [\![t_2]\!]_{M,g} \rangle \in I_M(A)$.
   By assumption it holds that $g(v) = g'(v)$ for every variable v that occurs in $At_1 \dots t_n$.
   So it holds that $[\![t_i]\!]_{M,g} = [\![t_i]\!]_{M,g'}$ for every t_i in $At_1 \dots t_n$.
2. $\phi = \neg\psi$:
   $V_{M,g}(\neg\psi) = 1$ iff $V_{M,g}(\psi) = 0$.
   From the induction hypothesis it follows that:
   $V_{M,g}(\psi) = 0$ iff $V_{M,g'}(\psi) = 0$ for all g, g' s.th. $g(v) = g'(v)$ for every variable v that occurs free in ψ.
   So it also holds that:
   $V_{M,g}(\neg\psi) = 1$ iff $V_{M,g'}(\psi) = 0$ for all such g, g' (clause $\neg$)
   and hence that $V_{M,g}(\neg\psi) = 1$ iff $V_{M,g'}(\neg\psi) = 1$ for all such g, g'.

Proof (cont.)

3. $\phi = \exists x \psi$:
   Consider g, g' s.th. $g(v) = g'(v)$ for every v that occurs free in $\exists x \psi$. The variables that occur free in ψ are the same as those that occur free in $\exists x \psi$ with the possible addition of x.
   From that it follows that $g[x/d](v) = g'[x/d](v)$ for every v that occurs free in ψ.
   From the induction hypothesis it follows that :
   $V_{M,g[x/d]}(\psi) = 1$ iff $V_{M,g'[x/d]}(\psi) = 1$,
   and thus that:
   $V_{M,g}(\exists x \psi) = 1$ iff $V_{M,g'}(\exists x \psi) = 1$ (clause $\exists$).

■

Consequence

Fact (2)

If ϕ is a sentence, then for all g, g':

$$V_{M,g}(\phi) = 1 \textit{ iff } V_{M,g'}(\phi) = 1$$

Substitution

Fact (3)

For all M and g:

$$V_{M,g}([c/x]\phi) = V_{M,g[x/I_M(c)]}(\phi)$$

The proof goes by way of induction on the construction of ϕ

Proof

Again we do three cases:

Proof:
Note that if x does not occur free in ϕ, $[c/x]\phi = \phi$ and the relevant result follows right away.

1. $\phi = At_1 \dots t_n$:
   Suppose $t_i = x$. We know that:
   $V_{M,g[x/I_M(c)]}(At_1 \dots x \dots t_n) = 1$ iff
   $\langle [\![t_1]\!]_{M,g[x/I_M(c)]} \dots [\![x]\!]_{M,g[x/I_M(c)]} \dots [\![t_n]\!]_{M,g[x/I_M(c)]} \rangle \in I_M(A)$,
   and that:
   $V_{M,g}(At_1 \dots c \dots t_n) = 1$ iff
   $\langle [\![t_1]\!]_{M,g} \dots [\![c]\!]_{M,g} \dots [\![t_n]\!]_{M,g} \rangle \in I_M(A)$.
   Since $g[x/I_M(c)](v) = g(v)$, for all $v \neq x$ and $g[x/I_M(c)](x) = I_M(c)$ the result follows immediately.
   Note that the case where x occurs more than once is not relevantly different.

Proof (cont.)

2. $\phi = \psi \wedge \chi$:
   Note that $[c/x](\psi \wedge \chi) = [c/x]\psi \wedge [c/x]\chi$.
   The induction hypothesis tells us that for all M and g:
   $V_{M,g}([c/x]\psi) = V_{M,g[x/I_M(c)]}(\psi)$,
   and the same holds for $[c/x]\chi$ and χ.
   On the basis of the truth definition (clause $\wedge$) the required result follows.

Proof (cont.)

3. $\phi = \forall y \psi$:
   Note that $[c/x]\forall y \psi = \forall y [c/x]\psi$.
   The induction hypothesis states that for all M and g:
   $V_{M,g}([c/x]\psi) = V_{M,g[x/I_M(c)]}(\psi)$.
   That means that for arbitrary d:
   $V_{M,g[y/d]}([c/x]\psi) = V_{M,g[x/I_M(c),y/d]}(\psi)$.
   And we have (clause $\forall$) the required result.

■

Generalisation

Can we extend Fact 3 to substitution by terms in general?

Extend notation:

- $[v'/v]\phi$ and $[t/v]\phi$, where v' a variable, and t a term.

Note:

- although x occurs free in $\exists y Rxy$, y in $[y/x]\exists y Rxy$ becomes bound: $[y/x]\exists y Rxy = \exists y Ryy$

Controlling substitution

Definition

y is free for x in ϕ iff there is no free occurrence of x in ϕ that is within the scope of an occurrence of $\exists y$ or $\forall y$

Let x, y, z, and w be different variables. Consider the formula :

$$\exists x Rxy \wedge (\forall y Ryx \vee Rzz)$$

Observe:

- y is not free for x: occurrence of x in $\forall y Ryx$ is free and in the scope of $\forall y$
- y is free for z: Rzz is not in the scope of $\forall y$
- w is free for x: no occurrences of $\exists w$ or $\forall w$
- x is not free for y: y is free in $\exists x Rxy$ and in the scope of $\exists x$

Generalisation of Fact 3

Fact (4)

For all M and g:

$$V_{M,g}[t/x]\phi = V_{M,g[x/[\![t]\!]_{M,g}]}\phi$$

provided t is free for x in ϕ

Proof: by induction on the construction of ϕ

Validity

Valid formulas

Definition (Universal validity)

A formula ϕ is (universally) valid iff for all M and g: $V_{M,g}(\phi) = 1$

Notation: $\models \phi$

Valid argument schemes

Definition (Validity of argument scheme)

An argument scheme $\phi_1 \ldots \phi_n \,/\, \psi$ is valid iff for all M and g:
if $V_{M,g}(\phi_1) = \ldots = V_{M,g}(\phi_n) = 1$, then $V_{M,g}(\psi) = 1$

Notation: $\phi_1 \ldots \phi_n \models \psi$

Logical equivalence

Definition (Logical equivalence)

ϕ and ψ are (logically) equivalent iff for all M and g:
$V_{M,g}(\phi) = V_{M,g}(\psi)$

Notation: $\phi \Leftrightarrow \psi$

Deduction theorem

Theorem (Deduction theorem)

1. $\phi \models \psi$ *iff* $\models \phi \to \psi$
2. $\phi_1 \dots \phi_n \models \psi$ *iff* $\phi_1 \dots \phi_{n-1} \models \phi_n \to \psi$

Repeated application of 2. delivers:

Theorem (Deduction theorem, general form)

$\phi_1 \dots \phi_n \models \psi$ *iff*
$\models \phi_1 \to (\phi_2 \to (\dots \to (\phi_n \to \psi) \dots))$ *iff*
$\models (\phi_1 \wedge \dots \wedge \phi_n) \to \psi$

Validity

Formulas

Reasoning about truth definition

Validity can be proven by reasoning about the truth definition.

Example:

$$\models (\forall x\phi \vee \forall x\psi) \to \forall x(\phi \vee \psi)$$

Proof. Choose arbitrary M and g and suppose $V_{M,g}(\forall x\phi) = 1$. Then it holds for all d: $V_{M,g[x/d]}(\phi) = 1$ (clause $\forall$). For arbitrary d it follows from $V_{M,g[x/d]}(\phi) = 1$ that $V_{M,g[x/d]}(\phi \vee \psi) = 1$ (clause $\vee$), which shows that $V_{M,g}(\forall x(\phi \vee \psi)) = 1$. ■

Using proven facts

In a proof we can make use of facts already proven. Example:

$$\models \forall x\phi \rightarrow [c/x]\phi$$

Proof. Suppose $V_{M,g}(\forall x\phi) = 1$ for arbitrary M and g. Then it holds for all $d \in D_M$: $V_{M,g[x/d]}(\phi) = 1$ (clause $\forall$). Since $I_M(c) \in D_M$, it follows that $V_{M,g[x/I_M(c)]}(\phi) = 1$, and on the basis of Fact 3 we can conclude that $V_{M,g}([c/x]\phi) = 1$. ■

Recall Fact 3:

$$V_{M,g}([c/x]\phi = V_{M,g[x/I_M(c)]}\phi$$

Validity

Argument schemes

Example

Example:

$$\exists y \forall x \phi \models \forall x \exists y \phi$$

Proof.

- Suppose for arbitrary M and g that $V_{M,g}(\exists y \forall x \phi) = 1$.
- Given the clause for $\exists$ this means that there is a d such that $V_{M,g[y/d]}(\forall x \phi) = 1$.
- With the clause for $\forall$ it follows that for d it holds that for every d': $V_{M,g[y/d,x/d']}(\phi) = 1$.
- Since $y \neq x$ it holds that $g[y/d, x/d'] = g[x/d', y/d]$.
- Then it follows from the clause for $\exists$ that for all d': $V_{M,g[x/d']}(\exists y \phi) = 1$,
- and with the clause for $\forall$ that $V_{M,g}(\forall x \exists y \phi) = 1$.

■

Validity

Equivalences

Logical equivalence

Definition (Logical equivalence)

ϕ and ψ are (logically) equivalent iff for all M and g:
$V_{M,g}(\phi) = V_{M,g}(\psi)$

Notation: $\phi \Leftrightarrow \psi$

Fact

Fact (5)

The following are all equivalent:

1. $\phi \models \psi$ *and* $\psi \models \phi$
2. $\phi \Leftrightarrow \psi$
3. $\models \phi \leftrightarrow \psi$

Useful equivalences

a. $\forall x \neg \phi \Leftrightarrow \neg \exists x \phi$

b. $\forall x \phi \Leftrightarrow \neg \exists x \neg \phi$

c. $\neg \forall x \phi \Leftrightarrow \exists x \neg \phi$

d. $\neg \forall x \neg \phi \Leftrightarrow \exists x \phi$

e. $\forall x (\phi \wedge \psi) \Leftrightarrow \forall x \phi \wedge \forall x \psi$

f. $\exists x (\phi \vee \psi) \Leftrightarrow \exists x \phi \vee \exists x \psi$

Useful equivalences (2)

g. $\forall x(\phi \vee \psi) \Leftrightarrow \phi \vee \forall x\psi$, if x not free in ϕ

h. $\exists x(\phi \wedge \psi) \Leftrightarrow \phi \wedge \exists x\psi$, if x not free in ϕ

i. $\forall x(\phi \rightarrow \psi) \Leftrightarrow \phi \rightarrow \forall x\psi$, if x not free in ϕ

j. $\forall x(\phi \rightarrow \psi) \Leftrightarrow \exists x\phi \rightarrow \psi$, if x not free in ψ

k. $\forall x \forall y \phi \Leftrightarrow \forall y \forall x \phi$

l. $\exists x \exists y \phi \Leftrightarrow \exists y \exists x \phi$

m. $\forall x \phi \Leftrightarrow \forall y [y/x]\phi$, provided y free for x in ϕ

n. $\exists x \phi \Leftrightarrow \exists y [y/x]\phi$, provided y free for x in ϕ

Using equivalences

Example:

$$\exists x(\phi \vee \psi) \Leftrightarrow \exists x\phi \vee \exists x\psi$$

Proof. $\exists x(\phi \vee \psi)$ is equivalent with $\neg\forall x\neg(\phi \vee \psi)$ (equivalence d.)
The latter is equivalent with $\neg\forall x(\neg\phi \wedge \neg\psi)$ (De Morgan)
That is equivalent with $\neg(\forall x\neg\phi \wedge \forall x\neg\psi)$ (equivalence e.)
That gives $\neg\forall x\neg\phi \vee \neg\forall x\neg\psi$ (De Morgan)
And that is equivalent with $\exists x\phi \vee \exists x\psi$ (equivalence d.) ■

Extensionality

Theorem (Extensionaliy)

$$\forall x_1 \ldots x_n(\phi \leftrightarrow \psi) \models \chi \leftrightarrow [\psi/\phi]\chi$$

where $x_1 \ldots x_n$ are the variables that occur free in ϕ and ψ

Using Extensionality

Simple example:

- Let $\phi = Px \wedge Qx$ and $\psi = Qx \wedge Px$
- We have that $\forall x((Px \wedge Qx) \leftrightarrow (Qx \wedge Px))$
- So we can use that to substitute:
  - Let $\xi = \exists x(Px \wedge Qx)$
  - Then we substitute $Qx \wedge Px$ for $Px \wedge Qx$
  - I.e.: $[(Qx \wedge Px)/(Px \wedge Qx)]\exists x(Px \wedge Qx)$
  - And obtain: $\exists x(Qx \wedge Px)$
- The Extensionality Theorem guarantees that the two are equivalent

Proving invalidity

Decidability

A set (a property) is *decidable* iff there is a mechanical procedure that will decide in a finite number of steps whether something belongs to that set (has that property)

Example: validity in proposition logic

Undecidability

Predicate logic is different:

Theorem (Church's theorem)

The set of valid formulas of predicate logic is not decidable

Given the Deduction Theorem this holds *a forteriori* for the set of valid argument schemes.

Counterexamples

A counterexample that shows that $\phi_1 \ldots \phi_n \not\models \psi$ is a model M and assignment g such that:

$$V_{M,g}(\phi_1) \ldots V_{M,g}(\phi_n) = 1 \text{ and } V_{M,g}(\psi) = 0$$

Example

Example:

$$\exists x(Ax \wedge Bx), \exists x(Bx \wedge Cx) \not\models \exists x(Ax \wedge Cx)$$

Proof. Consider the following model:

$$D = \{d_1, d_2\}, I(A) = \{d_1\}, I(B) = \{d_1, d2\}, I(C) = \{d_2\}$$

- Since $I(A) \cap I(C) = \emptyset$, the conclusion is false: $V_{M,g}(\exists x(Ax \wedge Cx) = 0$
- Since $d_1 \in I(A) \cap I(B)$, the first premise is true: $V_{M,g}(\exists x(Ax \wedge Bx) = 1$
- And since $d_2 \in I(B) \cap I(C)$, the second premise is true as well: $V_{M,g}(\exists x(Bx \wedge Cx) = 1$

■

Example (2)

Example:

$$\forall x \exists y Rxy, \forall x(Rxx \leftrightarrow Ax) \not\models \exists x Ax$$

Proof. Consider the following model:

$$D = \{d_1, d_2\}, I(R) = \{\langle d_1, d_2\rangle, \langle d_2, d_1\rangle\}, I(A) = \emptyset$$

- Since $I(A) = \emptyset$, we have that $V_{M,g}(\exists x Ax) = 0$
- Since there is no pair $\langle d, d\rangle$ in $I(R)$, we also have that $V_{M,g}(\forall x(Rxx \leftrightarrow Ax)) = 1$
- And for every d in D there is indeed a d' such that $\langle d, d'\rangle \in I(R)$: for d_1 this is d_2 and for d_2 it is d_1. So, $V_{M,g}(\forall x \exists y Rxy) = 1$

■

Constructing counterexamples

Note: we only need to specify the interpretation of predicates and constants that occur in the premisses or conclusion.

- Start with the minimal model:
  - $D = \{1\}$
  - $I(A), I(R), \cdots = \emptyset$
  - $I(c_i), \cdots = 1$
- Add elements to D and to $I(A), I(R), \ldots$ and change $I(c_1), \ldots$ in such a way that:
  - the premisses are made true
  - the conclusion remains false
- Use graphical representations of models for surveyability

Homework

Homework

- Read chapter 15 and make exercises 4, 6(a,c); 9(a,d,f)